STUDENT LABORATORY MANUAL

Module 7: Introduction to Computer Graphics and Visual Computing

20 Guided Examples • 20 Practice Exercises • 20 Outputs

Course: CS0045 – Computer Graphics and Visual Computing

Companion: Module 7 Student Guide

Toolchain: VS Code + MSYS2 UCRT64 GCC + GLEW + FreeGLUT

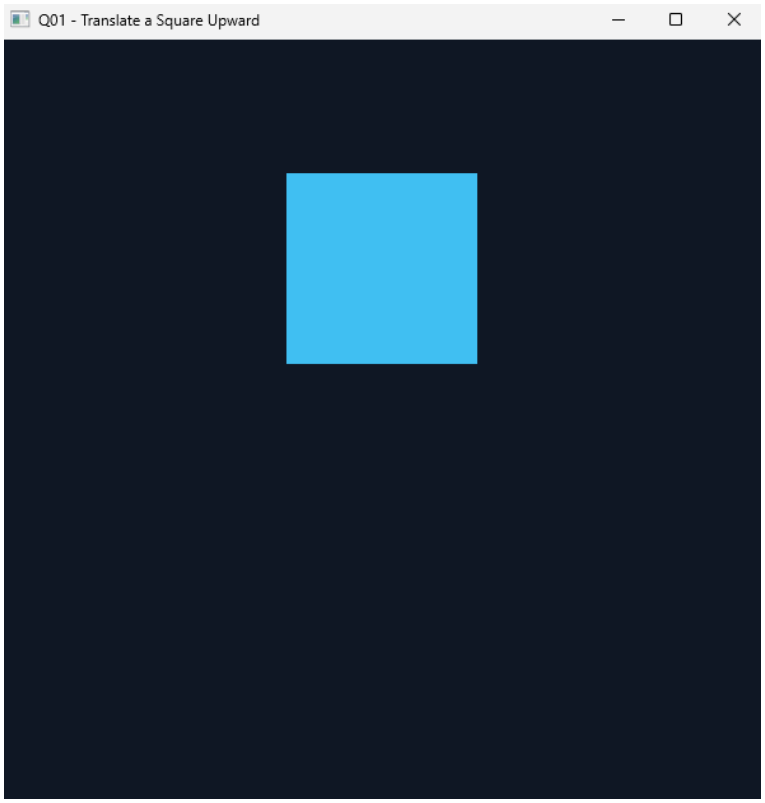
Student Name:	**Ebron, Nicole Anne V.**
Section:	**TN35**
Professor:	**Dr. Hadji J. Tejuco**

GitHub Repository: https://github.com/nicoleanneebron-art/Ebron-Nicole-Anne-V.

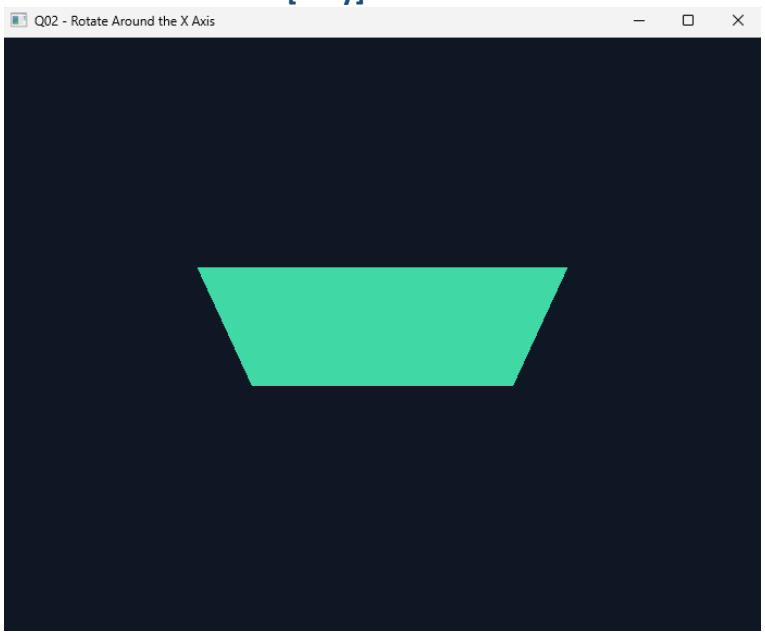
Part C — Outputs / Screen Shots (Ex01–Ex20)

Easy Exercises

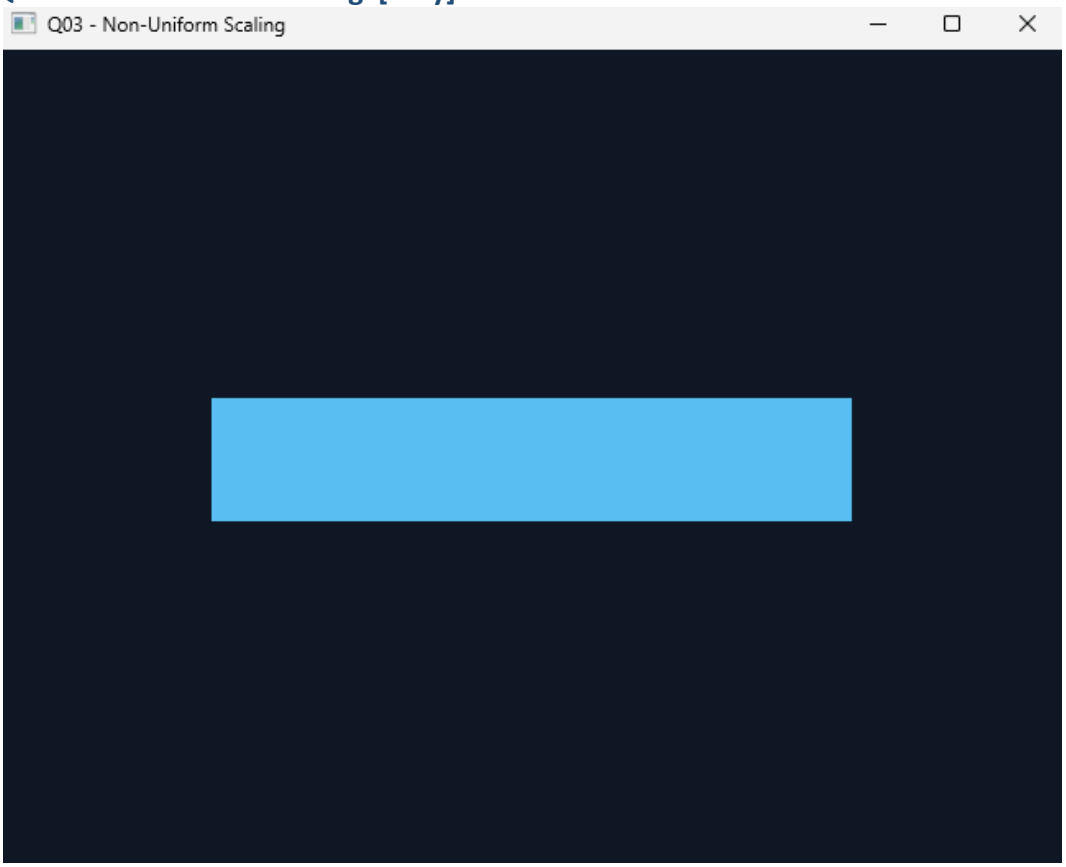
Exercise Q01 — Translate a Square Upward [Easy]



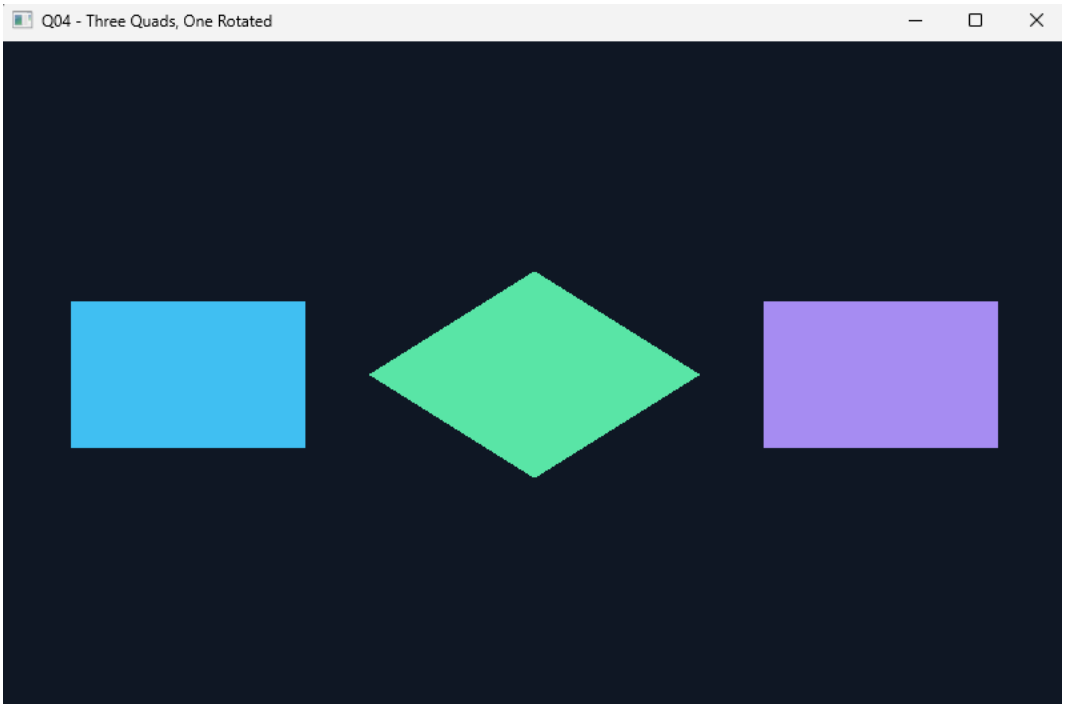
Exercise Q02 — Rotate Around the X Axis [Easy]



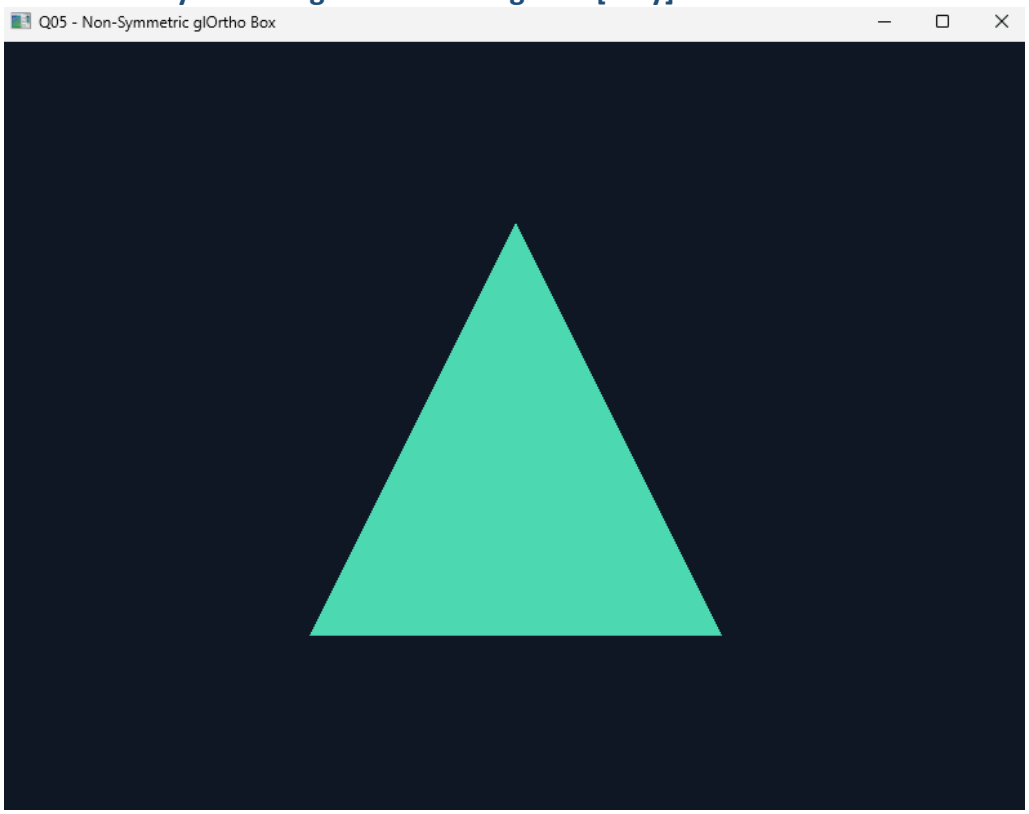
Exercise Q03 — Non-Uniform Scaling [Easy]



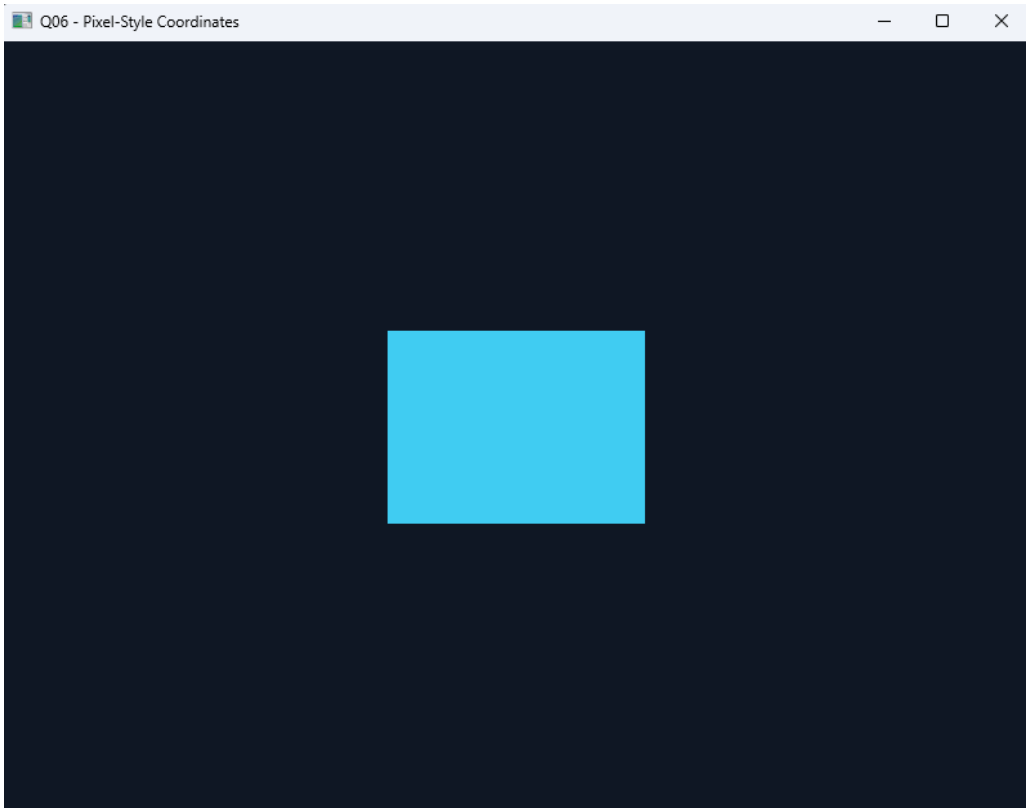
Exercise Q04 — Three Quads, One Rotated [Easy]



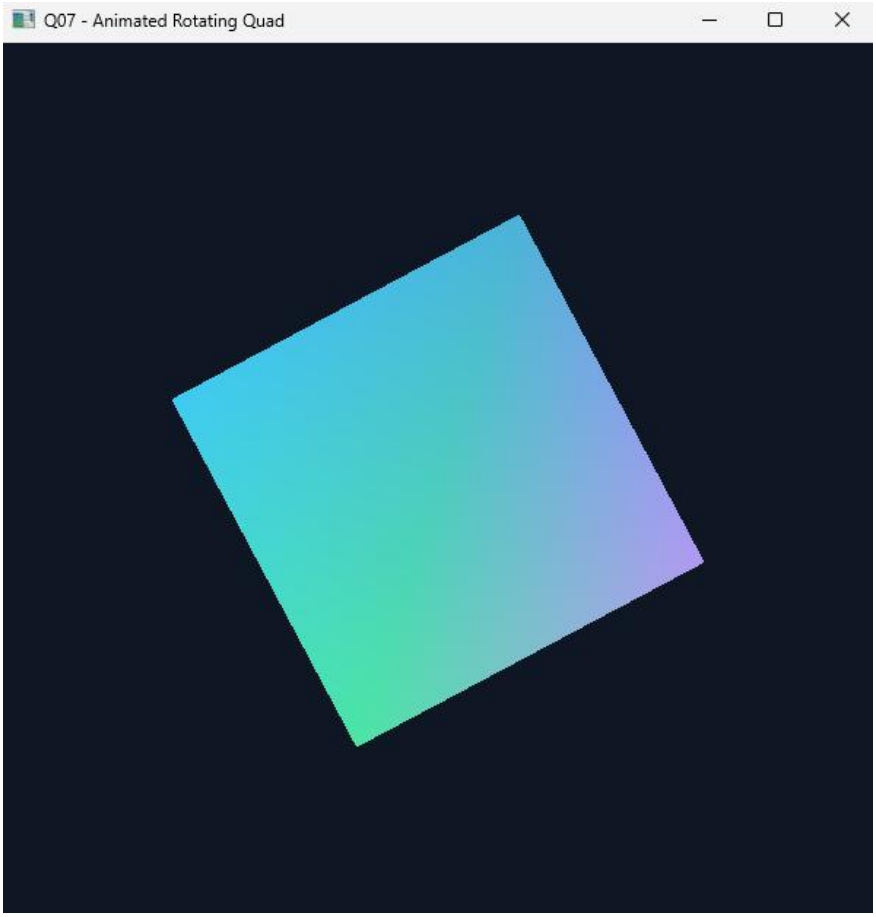
Exercise Q05 — A Non-Symmetric glOrtho Viewing Box [Easy]



Exercise Q06 — gluOrtho2D With Pixel-Style Coordinates [Easy]



Exercise Q07 — Animate a Rotating Quad [Easy]

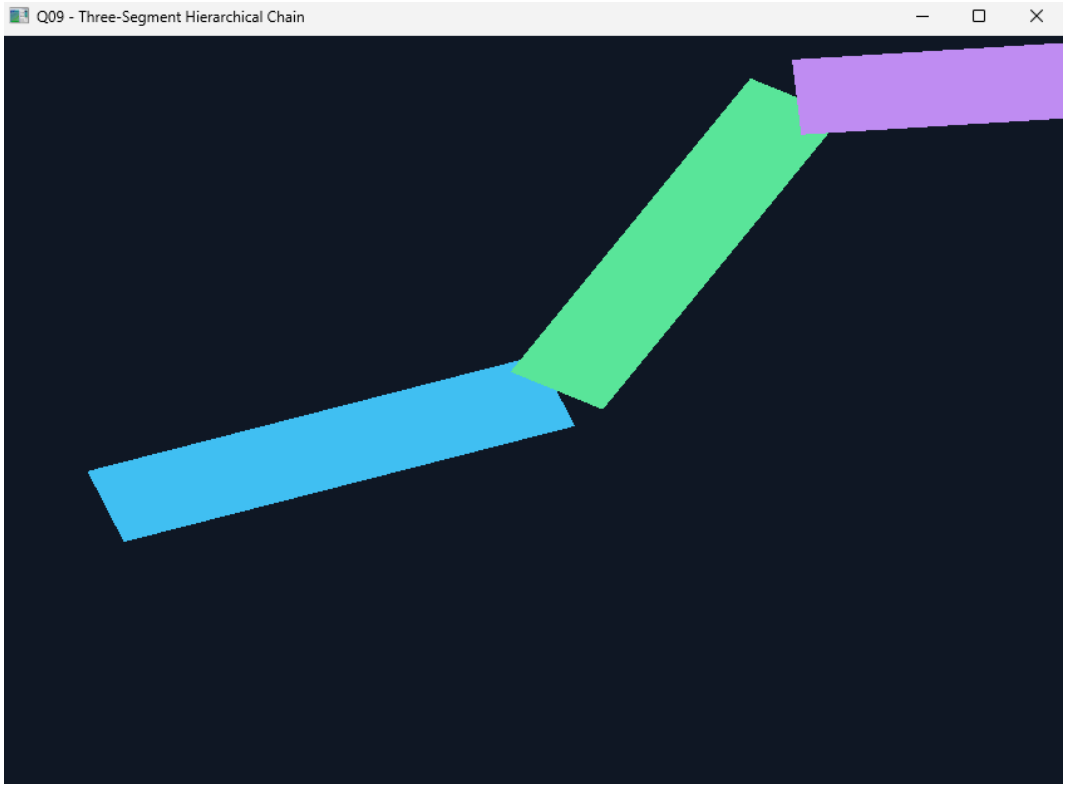


Medium Exercises

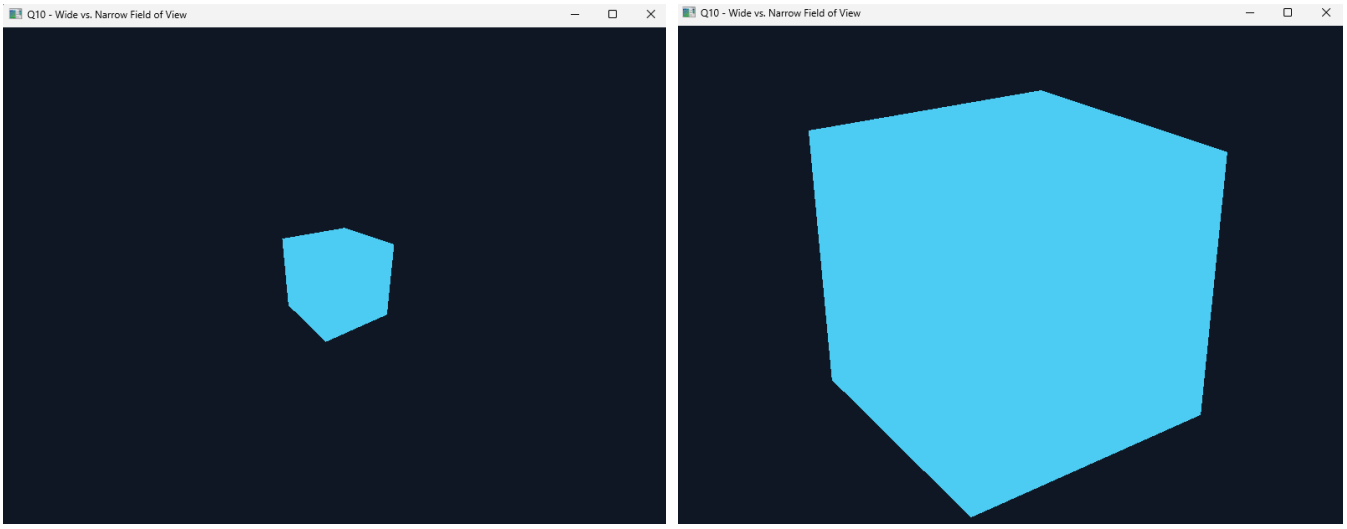
Exercise Q08 — Scale-Then-Translate vs. Translate-Then-Scale [Medium]



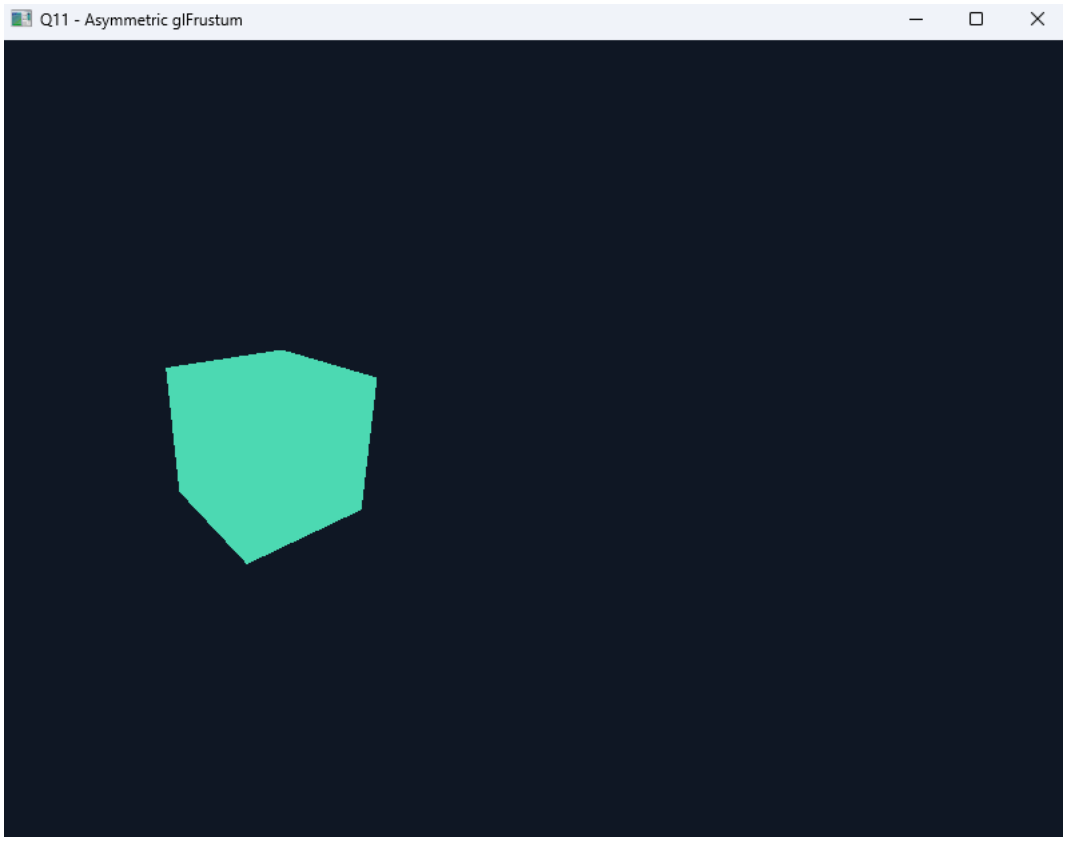
Exercise Q09 — Three-Segment Hierarchical Chain [Medium]



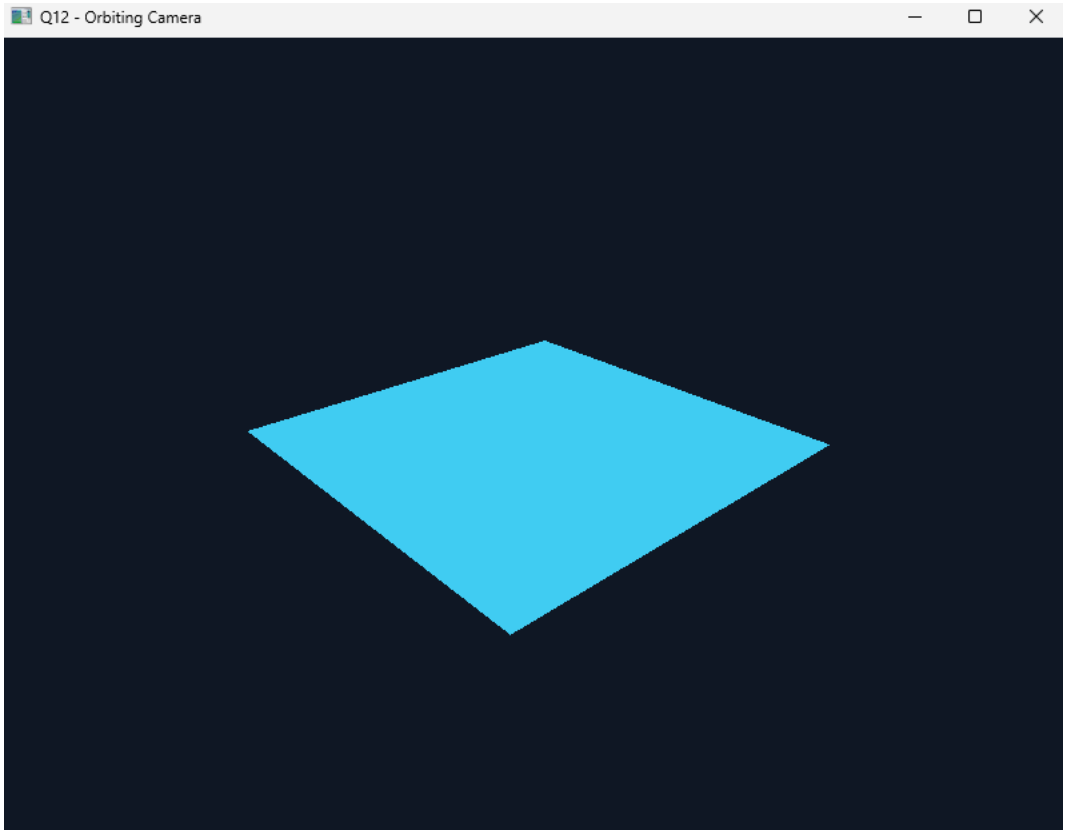
Exercise Q10 — Comparing Wide vs. Narrow Field of View [Medium]



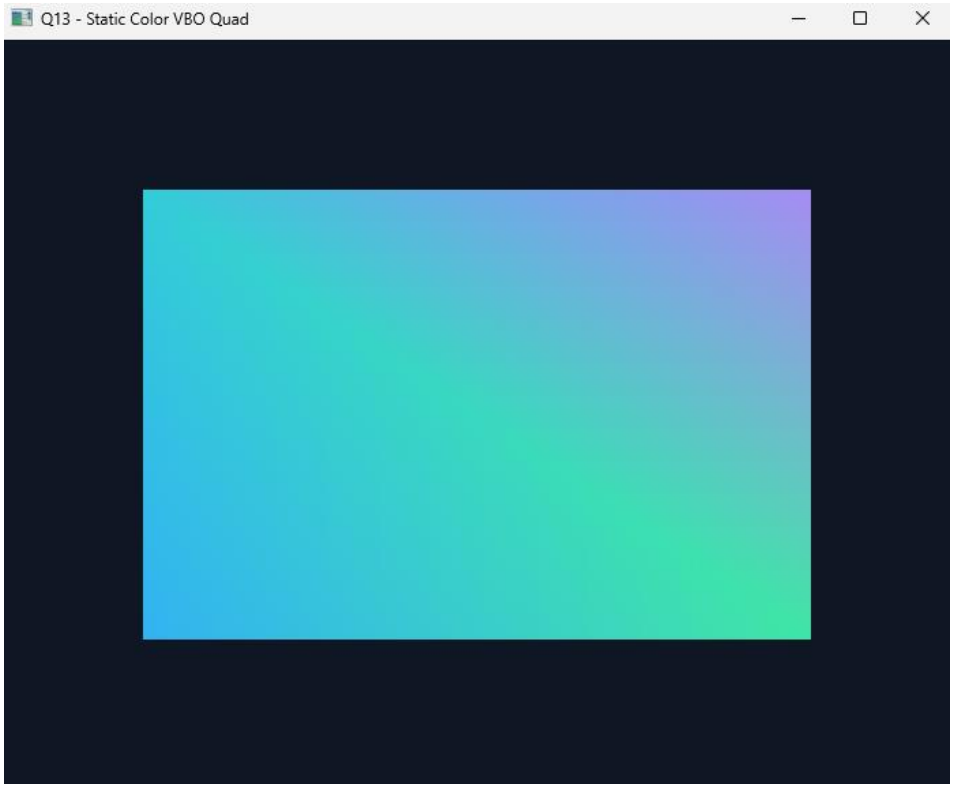
Exercise Q11 — An Asymmetric glFrustum [Medium]



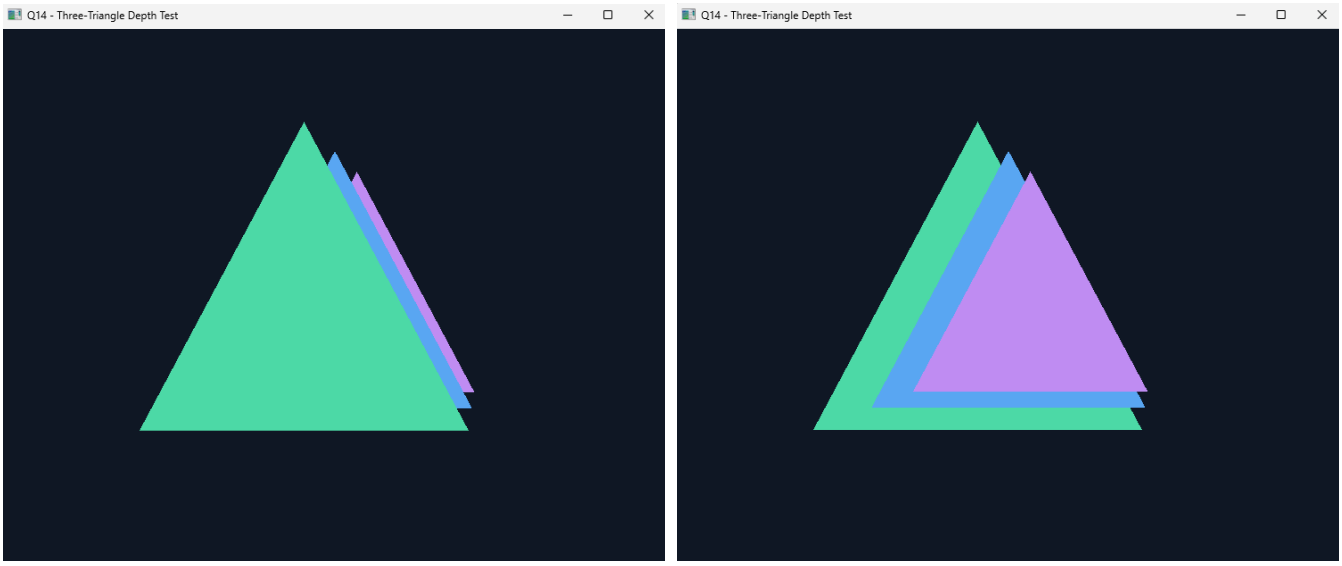
Exercise Q12 — Orbiting Camera via Animated gluLookAt [Medium]



Exercise Q13 — Static Color Quad via glBufferSubData + glArrayElement [Medium]

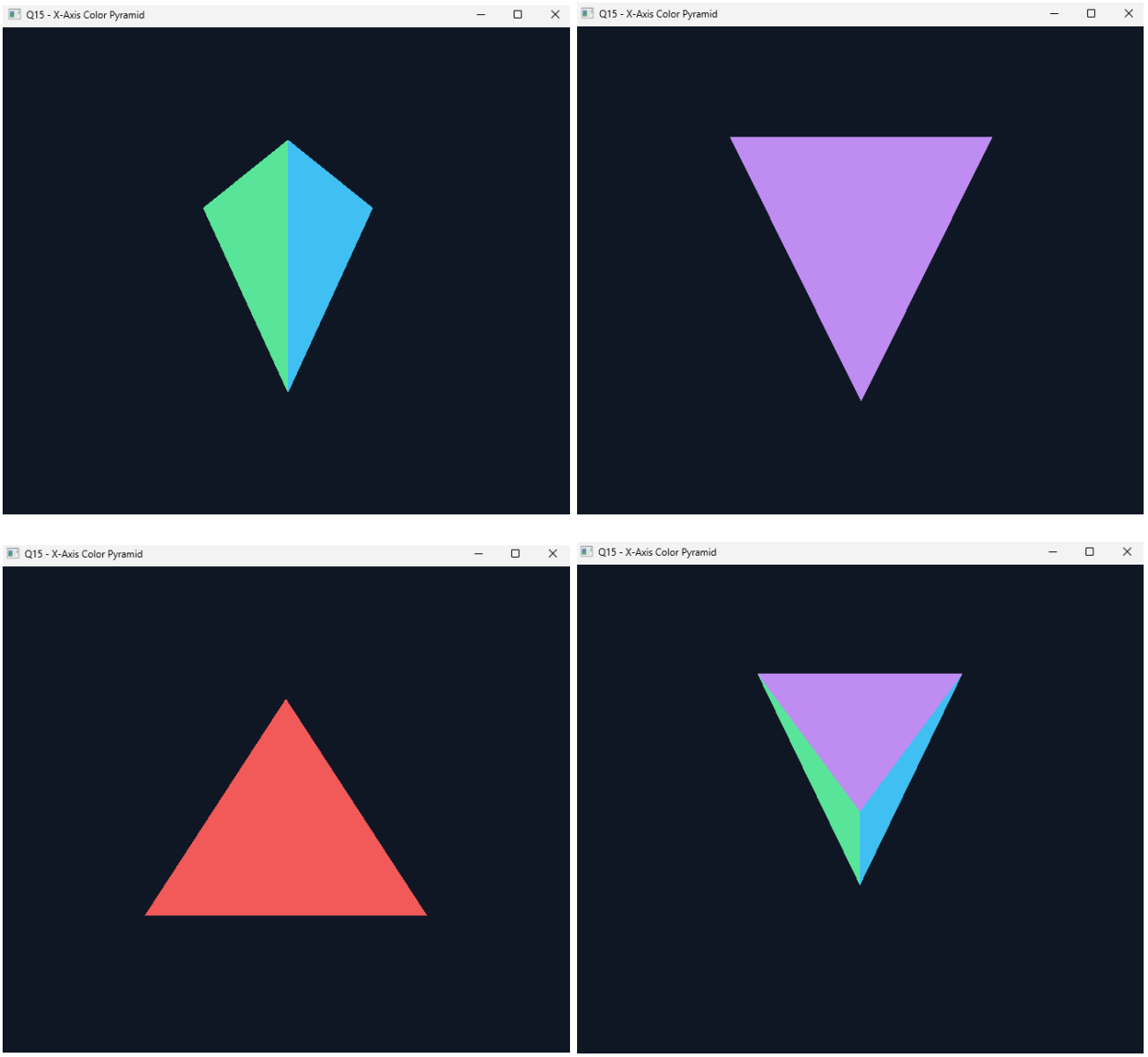


Exercise Q14 — Depth Testing With Three Triangles [Medium]

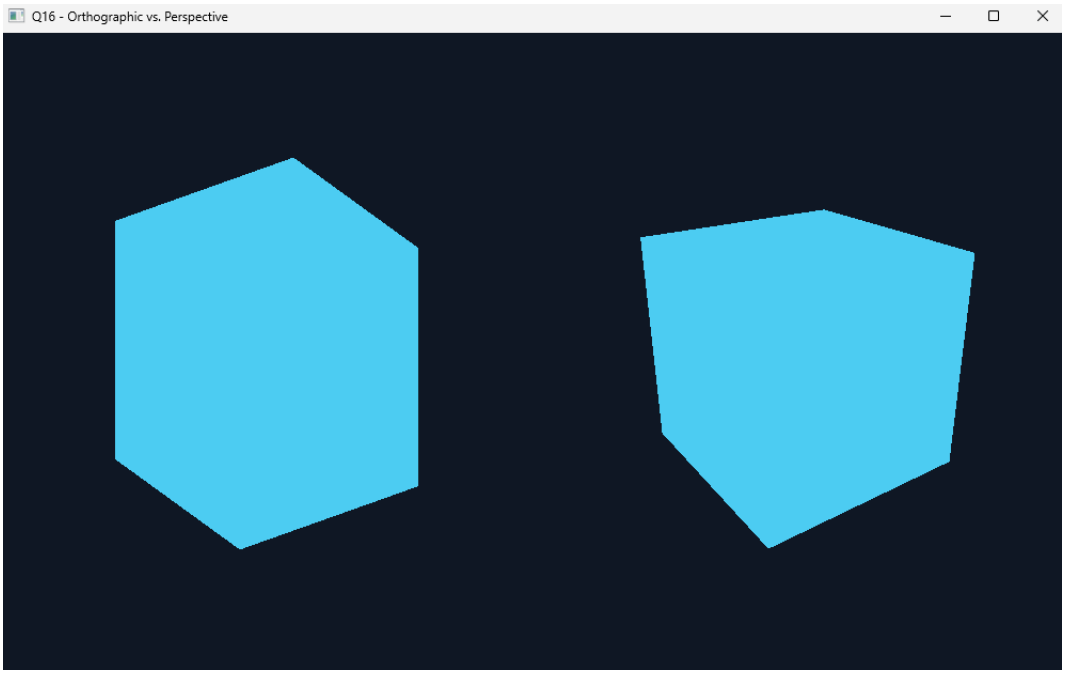


Hard Exercises

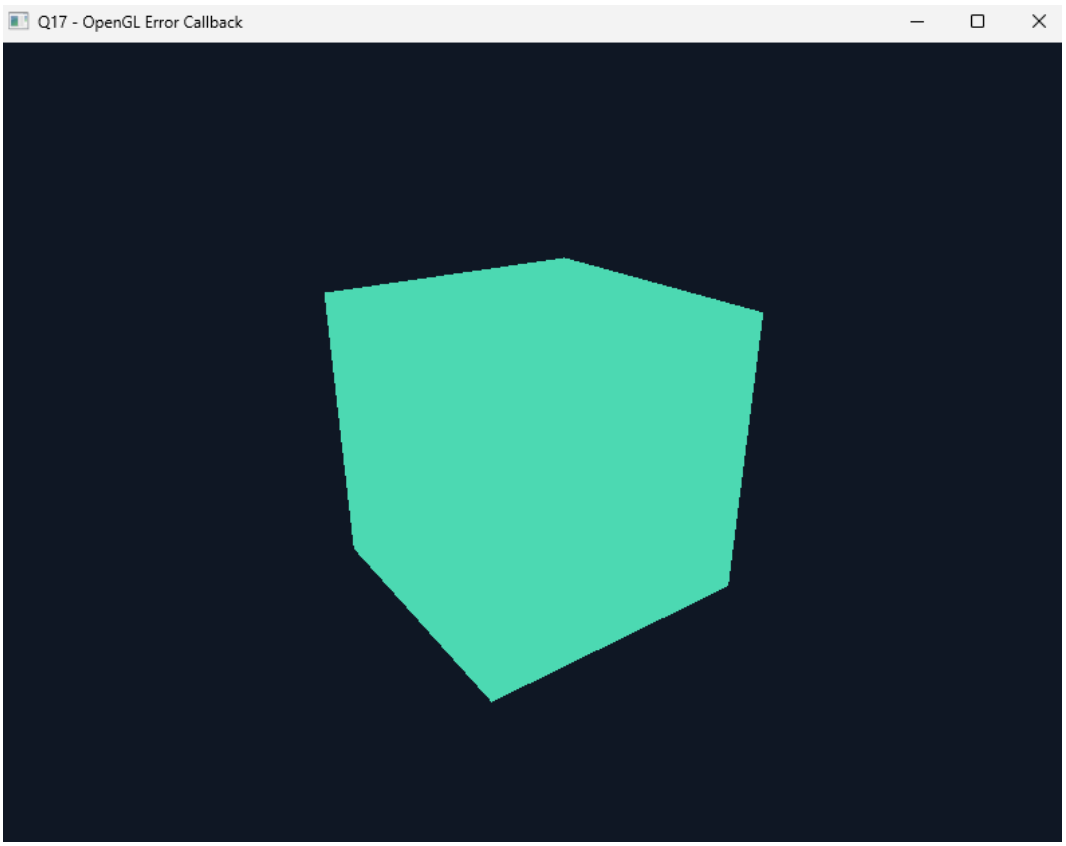
Exercise Q15 — Lecture Pyramid, Rotated on a Different Axis, With Cleanup [Hard]



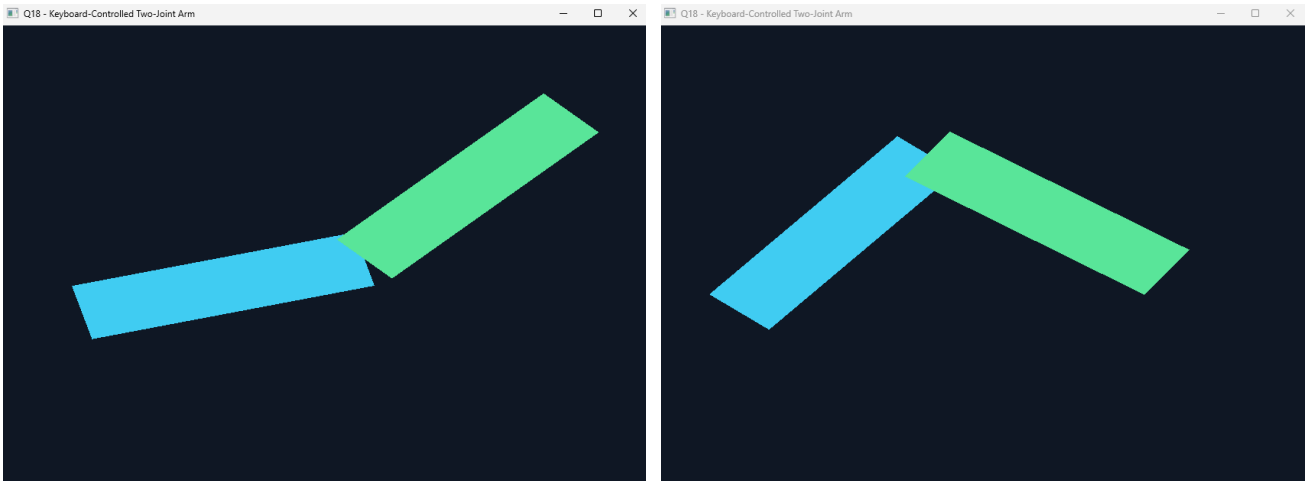
Exercise Q16 — Split Screen: Orthographic vs. Perspective, Side by Side [Hard]



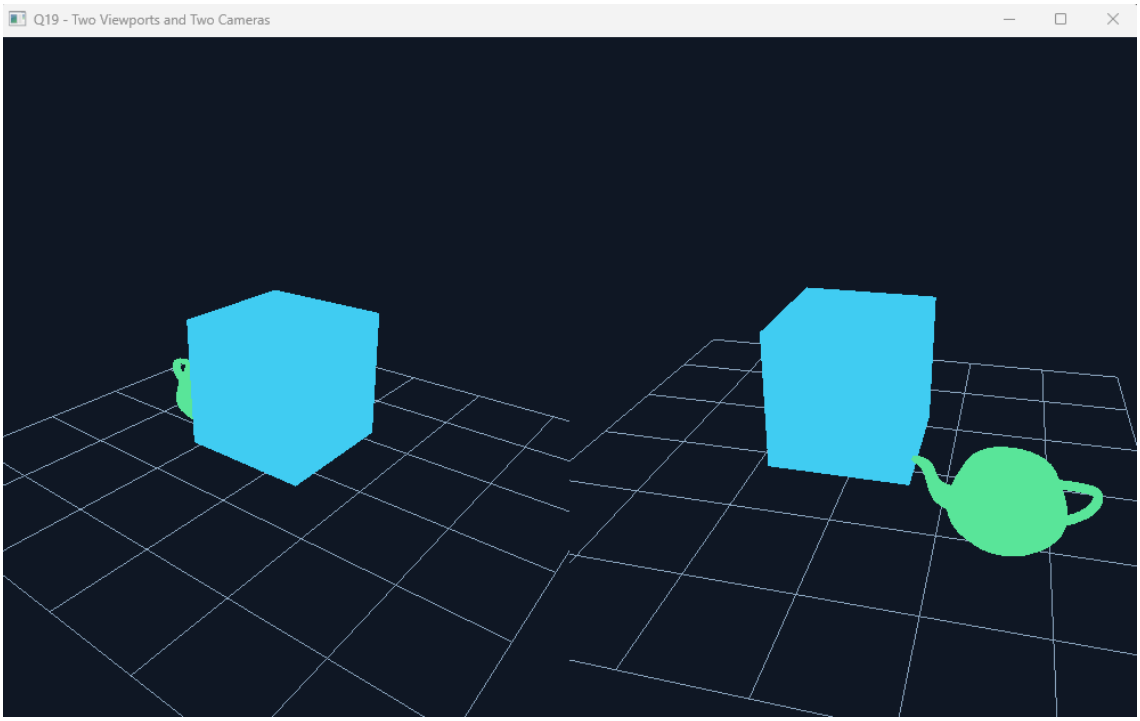
Exercise Q17 — Error Callback With an Intentional Trigger [Hard]



Exercise Q18 — Keyboard-Controlled Two-Joint Arm [Hard]



Exercise Q19 — Two Viewports, Two Cameras, One Animated [Hard]



Exercise Q20 — Capstone: Color VBO Base + Animated Hierarchical Arm [Hard]

